

XIAOYING LIAO

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Research Interests: AI for Biology; Computational Biology; Single-Cell & Spatial Genomics

INTRODUCTION

I am a researcher passionate about exploring the unknown and taking on challenging scientific questions. My research lies at the intersection of AI, computational biology, and experimental biology, integrating computational modeling with single-cell and spatial genomics to understand complex biological systems and uncover new biological insights.

EDUCATION

Johns Hopkins University (JHU)

Aug. 2024 – May 2025

Master of Science in Engineering (M.S.E.) in Biomedical Engineering

Baltimore, USA

- Supervised by Professor Carlo Iomini, **GPA:** 4.00/4.00
- Courses:** Cellular Engineering, Tissue Engineering, Cell & Tissue Engineering Lab, Advanced Topics in Regenerative and Immune Engineering, Human Genetics, Systems Pharmacology and Personalized Medicine

University of Sydney (USYD)

Feb. 2020 – Dec. 2023

Bachelor of Science (Honours) in Statistics

Sydney, Australia

- Supervised by Professor Jean Yang, **GPA:** 3.90/4.00, **Rank:** 1/200
- Courses:** Probability and Estimation Theory, Statistical Inference, Stochastic Process, Advanced Statistical Modelling, Statistical Machine Learning, Life and Evolution

PUBLICATIONS

Python Song, **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026. **Oral (Top 1%)**

Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 2026. pp. 3883.

Nianping Liu, Xuanzhi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.

Tianyang Wang, Xi Xiao, **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026. Under review.

Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.

Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smita Krishnaswamy, Jun Huan, Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.

Tianyang Wang, Xi Xiao, Gaoqi Chen, **Xiaoying Liao**, Guo Cheng, Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025. pp. 1124–1131.

Xiaoying Liao. *Assessing the Performance of Clustering Methods in Spatially Resolved Transcriptomics Data*. University of Sydney, 2023. Undergraduate honors research.

RESEARCH EXPERIENCE

Stanford University

2026–Present

Research Assistant, supervised by Prof. Xiaojie Qiu

Stanford, CA

- Human fetal heart development:** Analyzing PCW7 and 8 human fetal heart spatial transcriptomics data to characterize cardiac cell states, spatial niches, cell–cell communication, and 3D reconstruction during early heart development.
- 3D embryonic development:** Building a 3D spatiotemporal atlas of mouse embryogenesis from E5.5–E7.5 to characterize cell-state transitions, spatial tissue organization, and developmental trajectories during early embryogenesis.
- Pediatric cardiomyopathy:** Leading a single-cell multi-omics and spatial transcriptomics study comparing myocardial states before and after ventricular assist device support (VAD) and between recovery and non-recovery to identify cellular programs, spatial niches, and potential biomarkers associated with myocardial recovery.

Johns Hopkins University

2024–2026

Research Assistant, supervised by Prof. Carlo Iomini

Baltimore, MD

- Meibomian gland development:** Investigated how Hedgehog-responsive epithelial progenitors regulate gland development, meibocyte maturation, and lipid homeostasis.

- **Cilia–Hedgehog signaling:** Characterized Ift88–Smo genetic interactions and their effects on epithelial proliferation, gland morphology, and lipid production and composition.
- **Single-cell transcriptomics:** Generated and analyzed P5 and P10 meibomian gland scRNA-seq data to characterize developmental cell states and pathways associated with gland maturation and Hedgehog signaling.

University of Sydney

2023–2024

Research Assistant, supervised by Prof. Jean Yang and Prof. Clara Grazian

Sydney, Australia

- **Data preprocessing:** Used Python and R to preprocess spatial transcriptomics datasets, including quality control, normalization, feature selection, and dimensionality reduction using principal component analysis (PCA).
- **Spatial domain detection:** Applied and compared clustering approaches, including K-means and Seurat, alongside spatially informed methods such as BayesSpace and STitch3D, to identify tissue domains and characterize spatial organization.
- **Method evaluation:** Benchmarked clustering performance across simulated and real datasets, examining factors affecting method efficacy and identifying strengths and limitations to inform method selection in spatial transcriptomics research.

TEACHING EXPERIENCE

Johns Hopkins University

Jan. 2025 – May 2026

Teaching Assistant

Baltimore, United States

- **Cell & Tissue Engineering Lab:** Mentored students in practical modules on cell culture, gene delivery, metabolic glycoengineering, and tissue modeling. Provided support in data collection, data analysis, and scientific lab reports.

University of Sydney

Feb. 2023 – Dec. 2023

Teaching Assistant

Sydney, Australia

- **STAT2011 Probability and Estimation Theory:** Guided students through probability theories and statistical distributions, including practical R programming for data analysis applications.
- **DATA2002 Data Analytics: Learning from Data:** Led workshops on data analytics techniques, enabling students to apply statistical programming to real-world data challenges and machine learning concepts for informed decision-making.

PROFESSIONAL EXPERIENCE

JD.com

Feb. 2022 – May 2022

Data Analyst Intern

Beijing, China

- Analyzed large-scale retail and consumer datasets using SQL, Python, and R, applying time-series forecasting, regression, and statistical modeling.
- Developed automated analytics pipelines and dashboards to support inventory planning, consumer behavior analysis, and operational decision-making.

ByteDance – TikTok

Dec. 2021 – Feb. 2022

Data Operation Intern

Beijing, China

- Analyzed global business and user behavior data using SQL and Python, including automated metric monitoring and interactive BI dashboards.
- Applied statistical analysis and user segmentation models to characterize consumer behavior and support data-driven product and marketing strategies.

ACADEMIC ACTIVITIES

Stanford Genetics Department Retreat: Poster presentation	2026
Association for Research in Vision and Ophthalmology (ARVO): Poster presentation	2026
Annual Wilmer Research Meeting: Poster presentation	2025
Sydney Bioinformatics Research Symposium (SBRS): Participant	2023
Unlocking Single Cell Spatial Omics Analyses with scDney: Participant	2023

SKILLS & HONORS

Computational: Python, R, SQL, MATLAB, PyTorch, scikit-learn, Scanpy, Seurat; machine learning, statistical modeling, single-cell and spatial transcriptomics, multi-omics integration

Experimental: 10x Genomics single-cell sequencing, Illumina spatial transcriptomics, cell and tissue culture, immunostaining, Confocal and two-photon microscopy, PCR/qRT-PCR, Western blotting, DNA/RNA extraction

Honors: Vice Chancellor's Global Mobility Scholarship (2020); Dalyell Scholar (2022); 2nd Prize, China National High School Biology Olympiad (2018)